Hackathon Idea Guide

Every idea we looked at, explained in simple words

Team Hashes · Google Cloud AI Builder Cup 2026 · 11 October 2026

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1. The hackathon in one page

The **AI Builder Cup** is a competition run by Google Cloud for the Asia-Pacific region. Teams of 2 to 4 people build a working app that uses Google's AI (called **Gemini**) and runs on Google Cloud.

- **Deadline:** 18 October 2026 (we aim for 17 October).
- **What we hand in:** a live website link, public code on GitHub, a 3-minute video and a slide deck.
- **Pick one theme:** Banking, Retail, Manufacturing, Media, Future of Work, or Sustainability.
- **Prizes:** USD 30,000 in total. The winner gets USD 10,000. The final is in Singapore on 4 December.
- **Must be new:** nothing started before the program began.

Our current main plan is **ClearSight** (Sustainability theme). It checks whether companies keep the environmental promises written in their government approvals, using satellite images. This guide looks at other ideas in Retail, Media and Manufacturing, in case one of them is a better bet.

2. How judges score, and what our scores mean

What judges look at	Weight	In simple words
Technical merit and use of AI	40%	Does it work? Is the AI doing real work, not decoration?
Problem fit and impact	25%	Does it solve a real, painful problem for real people?
Innovation and creativity	25%	Is it new? Is the AI used in a clever way?
User experience	10%	Is it easy and pleasant to use?

Our score is our own rough guess, from 1 to 5, using the same weights. It is not official.

Score	What it means
4.5 or more	Very strong. Only one small weakness.
4.0 to 4.4	Good, but one clear weakness (someone already does part of it, or the data is thin).
3.5 to 3.9	Weak. Mostly done by others, or hard to build in 6 days.
below 3.5	Drop it.

A difference of less than 0.3 between two ideas is basically a tie. How well we build and present the idea matters more than these numbers.

Words we use for competition: Open = nobody does it yet. **Partly taken** = someone does part of it, so we must show what is different. **Taken** = it already exists; avoid.

3. Word list: hard terms in plain English

Term	Plain meaning
Gemini	Google's AI. It can read text, look at pictures, watch videos and listen to audio.
Generative AI (Gen AI)	AI that creates new things: text, images, audio, plans.
Human in the loop	The AI suggests, but a person checks and makes the final decision.
Ground truth	The known correct answers we use to test how accurate our AI is.
Precision / recall	Precision: of the things the AI flagged, how many were right. Recall: of all real problems, how many the AI found.
Cloud Run	Google's service that runs our app on the internet.
Changeover	Stopping a factory machine to switch it from making one product to another (for example, swapping a mould).

Term	Plain meaning
SMED	A Toyota method to make changeovers much faster, mainly by doing jobs before the machine stops.
Recall	When a government or company orders a dangerous product off the market.
Marketplace	A big shopping website where many sellers list products (Amazon, Flipkart, Shopee).
Listing	A product's page on a shopping website: title, photos, price, details.
Legal Metrology	Indian law that says what must be printed on a pack and shown online: price (MRP), weight, origin, expiry date, maker.
CCPA	India's consumer protection regulator. It sends notices and fines to shopping sites.
BIS / ISI mark	India's product safety certification. Many electronics must carry it.
ACCC	Australia's consumer and product-safety regulator.
Decodable book	A children's reading book that only uses letters the child has already learned.
Akshara	A letter or syllable sign in Indian scripts like Hindi or Marathi.
LQA	Localization quality assurance: checking that a translated game or app looks and reads right.
NER score	A standard way to measure how accurate live TV captions are (98 out of 100 is a pass in Canada).
Mystery shopper	A person who pretends to be a customer to check how good a store's service is.
Visual merchandising (VM)	How products and posters are arranged in a store, following the brand's rules.
ONDC	India's government-backed open network that lets small shops sell online.
Entailment check	An automatic check that a sentence really is supported by its source.

4. What other teams are already building

We searched GitHub five times for public projects that name the AI Builder Cup. We found about **125 real entries** (many more teams are still private until the deadline).

Theme	Public entries	What most of them do
Sustainability	~30	Disaster response, farming, climate, accessibility
Banking (BFSI)	~29	Scam and fraud detection (very crowded)
Future of Work	~26	Meeting tools, AI agents, hiring, documents
Retail	~16	Shop assistants, inventory, product search, scam checks
Manufacturing	~14	Machine maintenance, root-cause analysis, quality checks
Media	~10	Audio description, ad localization, learning videos

What this tells us: Retail and Media have fewer teams, so there is more room to stand out. None of the ideas in this guide is being built by another public team.

5. All ideas at a glance

#	Idea	Theme	Score	One line
-	ClearSight (current plan)	Sustainability	4.45	Checks companies' environmental promises with satellite images
-	PitStop	Manufacturing	4.39	Phone video of a machine changeover turns into a faster plan
1	SafeShelf	Retail	4.45	Spots dangerous products from buyer reviews before the recall
2	Akshara Ladder	Media	4.45	Makes Hindi/Marathi reading books using only letters a child knows
3	ShopTest	Retail	4.38	AI mystery shopper that also trains the staff who made mistakes
4	LQA Lens	Media	4.33	Watches game videos to find translation mistakes
5	ImageHonest	Retail	4.25	Checks if product photos match what buyers actually received
6	Sansad Clips	Media	4.25	Turns long parliament debates into short, fair, local-language clips
7	ShopReel	Retail	4.13	One video of a small shop becomes a full online catalog
8	VM Fix	Retail	4.13	Checks store displays against brand rules and shows the fixed version
9	CaptionAudit	Media	4.08	Scores how accurate live TV captions are
10	PackTruth	Retail	4.04	Checks product pages against what is printed on the real pack
11	Comic Voices	Media	4.00	Turns comics into audio dramas for blind readers
12	Radio Rewind	Media	4.00	Turns old community-radio recordings into new local bulletins

Honest note: after checking competitors, no Retail or Media idea scored a clear 4.5. Two ideas (SafeShelf and Akshara Ladder) reach about 4.45 and could reach 4.5 if one key test works.

6. Manufacturing idea

PitStop

Theme: Manufacturing **Score:** about 4.39

Like an F1 pit crew: film a machine changeover on a phone, and the AI shows how to do it much faster.

The problem, in plain words

When a factory switches a machine from one product to another, the machine stands idle. This can take 45 minutes, several times a day. Toyota's SMED method can cut this time in half, but it needs an expert with a stopwatch for days. Most small factories cannot afford that. In India the government even pays 90% of consultant fees for this.

What we would build

- The engineer films one changeover with a phone and talks while working ("machine stopped", "removing mould").
- Gemini watches the video and breaks it into steps with start and end times.
- For each step it says: does the machine really need to be stopped for this, or could it be done earlier?
- The engineer fixes any wrong steps on a simple timeline screen.
- Gemini suggests a new plan (do these steps early, give that one to a second person). The computer calculates the new time, for example 45 to 26 minutes.
- It prints a one-page instruction sheet with pictures from the video, in Hindi, Thai or other local languages.
- The next changeover is filmed too, and PitStop shows how much time was really saved.

Who uses it

Ravi, the only process engineer at a 120-person plastic-parts factory in Pune.

Who already does something similar

Some tools (Coats, MTM, Retrocausal) already time short, repeated factory tasks from video. Japanese tools need training on repeated work. None of them handles a long, one-time changeover from a phone video and then checks the saving with a second video. No other team in the cup does changeovers.

Why it could win

Easy story (F1 pit stops), clear before-and-after numbers, and the AI does real work (watching long videos).

Biggest risk

The AI may get step start and end times a bit wrong. Talking while filming and the correction screen fix most of this.

7. Retail ideas

1. SafeShelf

Theme: Retail **Score:** about 4.45 (could reach 4.5)

Finds dangerous products on shopping sites from buyer complaints and photos, before the official recall.

The problem, in plain words

Shopping sites keep selling dangerous products: chargers that melt, toys with loose magnets, fake safety marks. An OECD check found 87% of banned or recalled products were still for sale online. Usually buyers complain and post photos weeks before anyone acts. Australia's regulator took Amazon to court in 2026, and India raided Amazon and Flipkart warehouses over fake ISI marks.

What we would build

- Gemini reads reviews and looks at buyer photos (burn marks, melted plastic, broken parts).
- It groups the same product even when it is sold under different brand names, by comparing photos.
- It reads safety marks and numbers on the listing and checks them against official government lists.
- It prepares an evidence pack and a draft report for the regulator.
- A safety analyst checks everything and decides.

Who uses it

A product-safety analyst at a shopping website, or at a regulator.

Who already does something similar

DeepRecall removes products only after the official recall. Research tools read review text only, not photos. Nobody combines photo evidence, fake-mark checks and early warning.

Why it could win

Saves lives, uses Gemini's image reading in a serious way, and we can prove it works with past data.

Biggest risk

Some government registries are hard to access automatically. Australia's list is open and can be used instead.

What would make it stronger

Test it on 20+ past US recalls using old Amazon reviews. If it raises the alarm at least 2 weeks early, that is the headline of the demo.

3. ShopTest

Theme: Retail **Score:** about 4.38

An AI mystery shopper that calls stores, checks the staff's answers, and then trains them.

The problem, in plain words

Big store chains pay people to pretend to be customers and check service quality. It is slow and expensive (a 2.8 billion dollar market). One study of 10,000 calls found 3 in 10 calls were not answered and almost nobody called back. When staff give wrong answers, nobody trains them on that exact mistake.

What we would build

- Gemini Live phones the store or chats on WhatsApp as a realistic customer, in local language (for example Taglish).
- It compares the staff's answers with the store's own rule book.
- It gives a score with proof ("at 1:42 the staff gave the wrong return policy").
- The manager approves the score.
- The AI creates a short practice role-play for that staff member on the exact mistake.

Who uses it

Maria, area manager of 40 eyewear and electronics stores in Manila.

Who already does something similar

AI training tools exist (Second Nature, YOOBIC) but do not check real stores. Call-scoring exists for car dealers. Nobody joins checking and training together for Asian stores.

Why it could win

Live, fun demo, and it closes the loop from finding a problem to fixing it.

Biggest risk

Recording real store staff without telling them raises consent issues. The demo uses our own team as staff.

5. ImageHonest

Theme: Retail **Score:** about 4.25

Checks whether product photos on a listing match what buyers actually received.

The problem, in plain words

Many sellers use edited or AI-made photos that make products look better than they are. Buyers feel cheated and return items. Tools exist to spot AI images, but not to check if the product looks the same in real life.

What we would build

- Compare the seller's photos with photos buyers posted in reviews.
- Check each feature: colour, size, material, number of pieces.
- Show the pictures side by side with the differences marked.
- Give sellers a pre-check before their listing gets flagged.

Who uses it

A trust and safety analyst at a shopping website.

Who already does something similar

Fake-image detectors and guides exist. Fakespot closed in 2025. Nobody compares feature by feature.

Why it could win

Very visual demo and a new problem caused by AI photos.

Biggest risk

Hard to get correct answers for testing. We would need to label 200 photo pairs by hand.

7. ShopReel

Theme: Retail **Score:** about 4.13

A small shop owner films one video walking through the shop, and gets a full online catalog.

The problem, in plain words

Over 11 lakh small Indian shops have joined ONDC, but most never finish their online catalog because typing every product is slow.

What we would build

- The owner films the shop while saying prices out loud.
- Gemini finds each product, links the spoken price to it, and cuts out a clean photo.
- It writes descriptions in two languages and fills in sizes and variants.
- It asks the owner on WhatsApp only about items it is unsure of.
- It exports the catalog to ONDC.

Who uses it

A small kirana or clothing shop owner.

Who already does something similar

Partly done: Google showed a photo-to-catalog demo for ONDC, Amazon makes listings from one photo, and a Gemini competition project did a business video. Linking spoken prices to many products in one video is the new part.

Why it could win

Huge number of users and a great "90 seconds to 40 products" moment.

Biggest risk

Judges may think it has been done before.

8. VM Fix

Theme: Retail **Score:** about 4.13

Checks store displays against the brand's rule book, then shows a picture of how it should look.

The problem, in plain words

Brands send franchise stores thick guideline PDFs on how to arrange displays. Only about 4% of store programs are carried out fully.

What we would build

- Read the brand guideline PDF and turn it into a checklist automatically.
- Look at a store photo or video and list what is wrong.
- Create an edited image showing the correct display.
- The store takes a new photo, and the tool confirms it is fixed.

Who uses it

Franchise operations lead at a bubble-tea chain in Jakarta.

Who already does something similar

Crowded: YOOBIC, Trax, ParallelDots and others. The new parts are zero setup and the generated "fixed" image.

Why it could win

Nice before-and-after pictures.

Biggest risk

Many competitors, so it is hard to look new.

10. PackTruth

Theme: Retail **Score:** about 4.04

Checks product pages on shopping sites against what is printed on the real pack.

The problem, in plain words

Indian law says the price, weight, origin and expiry date must be correct online. The regulator sent notices to 11 big shopping and quick-delivery companies, and almost half of shoppers cannot find expiry dates online.

What we would build

- Read all sides of the pack photo (Hindi or English).
- Compare each required detail with the website listing.
- Show the wrong detail, the photo crop and the rule broken.
- Suggest the correct text and create a reply pack for the regulator.

Who uses it

Priya, compliance lead at a quick-delivery app like Blinkit or Zepto.

Who already does something similar

Woflow does something close in the US. Label tools check artwork before printing, not live listings.

Why it could win

Real government pressure and easy test data (Open Food Facts).

Biggest risk

Judges may see it as just reading text and comparing.

8. Media ideas

2. Akshara Ladder

Theme: Media (publishing and education) **Score:** about 4.45 (could reach 4.5)

Makes children's reading books in Hindi or Marathi that use only the letters a child has already learned.

The problem, in plain words

Only about 23 out of 100 children in Class 3 in India can read a Class 2 text. Children learn to read faster with "decodable" books, which use only known letters. Teachers in India do not have these books for each week of teaching. AI tools make them only in English.

What we would build

- The teacher picks the letters taught so far ("week 3").
- Gemini rewrites a free story from StoryWeaver using only those letters.
- A computer checker counts every letter and sends it back to Gemini until at least 95% of words are readable.
- It adds matching pictures and a read-aloud voice.
- The teacher checks and approves the book.

Who uses it

A Class 1 government-school teacher in Maharashtra.

Who already does something similar

LitLab and Project Read make decodable books, but only in English. Gemini Storybook makes books but does not limit letters. We found no tool for Indian scripts.

Why it could win

Big social impact, clearly new, and easy to measure (percent of readable words).

Biggest risk

Judges may call it Education rather than Media. We present it as a children's book publishing tool.

What would make it stronger

Show at least 95% readable words, against under 50% for plain Gemini, using the real letter order from a state syllabus.

4. LQA Lens

Theme: Media **Score:** about 4.33

Watches recorded gameplay and finds translation mistakes in games.

The problem, in plain words

When a game is translated into Japanese, Korean or Thai, testers play for hours looking for cut-off text, untranslated words or wrong politeness levels. This costs 41 to 65 dollars per hour. An industry analyst said in 2025 that no full solution exists.

What we would build

- Gemini watches 30 to 40 minutes of gameplay video.
- It flags cut-off text, missing translations, broken words and cultural problems.
- It links each problem to the exact text line in the game files.
- It suggests a translation that fits the space.
- A language expert approves before a ticket is made.

Who uses it

The translation testing lead at a game publisher in Tokyo.

Who already does something similar

Gridly checks text only. Game testing tools check gameplay, not language. Research looks at visual glitches only.

Why it could win

Clearly open gap and strong use of long-video AI.

Biggest risk

Only interesting to game companies, so the impact may look small.

6. Sansad Clips

Theme: Media **Score:** about 4.25

Turns long parliament debates into short, fair video explainers in local languages.

The problem, in plain words

Parliament debates last hours, and many laws pass with very little discussion. Most citizens never see what was said, especially in their own language.

What we would build

- Gemini watches the full debate video.
- It writes a 90-second explainer for each law, in Tamil or Hindi.
- Every sentence links to the exact second in the original video.
- An automatic check confirms each sentence matches what was actually said.
- It makes sure both government and opposition voices are included. An editor approves.

Who uses it

An explainer editor at Sansad TV or a regional news channel.

Who already does something similar

US tools (Hamlet, citymeetings.nyc) summarise city meetings. India's Sansad Bhashini translates but does not make explainers.

Why it could win

Strong for democracy and every claim can be checked.

Biggest risk

Political topics are sensitive. The tool must clearly show it is neutral.

9. CaptionAudit

Theme: Media **Score:** about 4.08

Scores how accurate live TV captions are, for deaf and hard-of-hearing viewers.

The problem, in plain words

Live captions often have mistakes, like "15%" shown as "50%", or they cover the score on screen. Checking accuracy by hand takes about 2 hours for 10 minutes of TV. Australia's regulator found several channels broke caption rules.

What we would build

- Upload the TV recording and the caption file.
- Gemini compares what was said with what was captioned.
- It grades each mistake (small, normal, serious) and checks if captions block important pictures.
- It produces a score and a report in the regulator's format.
- An expert reviews the flagged mistakes.

Who uses it

Caption compliance manager at a regional TV network in Australia.

Who already does something similar

A university tool (NER Buddy) and caption vendors check their own work. None checks the screen picture too.

Why it could win

Helps accessibility with a clear score.

Biggest risk

Small number of users, and we cannot freely use TV recordings.

11. Comic Voices

Theme: Media **Score:** about 4.00

Turns manga and webtoons into multi-voice audio dramas for blind readers.

The problem, in plain words

Blind and low-vision people cannot enjoy comics, which are very popular in Asia.

What we would build

- Read each panel in the right order (manga is read right to left).
- Describe the scene and give each character a voice.
- The creator approves the script before audio is made.

Who uses it

A comics publisher's accessibility team.

Who already does something similar

ComicCast (Spain, with Amazon) already does this in Spanish. Asian reading order and Indian languages are open.

Why it could win

Emotional demo.

Biggest risk

Partly done already, and the best test dataset needs an access request.

12. Radio Rewind

Theme: Media **Score:** about 4.00

Turns old community-radio recordings into new bulletins in local languages.

The problem, in plain words

Community radio stations have years of useful recordings (farming tips, health advice) that nobody can search.

What we would build

- Make the archive searchable by topic.
- On request ("all monsoon crop tips from 2024"), find clips.
- Write and voice a new 3-minute bulletin.
- The station editor approves.

Who uses it

A community radio manager in Meghalaya.

Who already does something similar

US broadcast search tools exist. Nobody turns archives into new local-language programs.

Why it could win

Helps rural listeners.

Biggest risk

Gemini may not understand rare languages like Khasi well.

9. Ideas we checked and threw away

These were checked and dropped, mostly because someone already does them.

Idea	Why we dropped it
Recall sweep (recalled products still sold)	DeepRecall and Recall Shadow already do it
Returns reason analysis	Clootrack, Enterpret and others
Shrinkflation tracker	Frugl does it in Australia
Fake discount checker	Mostly simple maths; real price data hard to get
Checkout overcharge detector	AI part is shallow
Shop display scheme payouts	SalesCode already does it
Out-of-stock substitutions	Instacart, Pick n Pay (on Gemini)
Complaint-based defect warning	SentiSum and others
Supplier onboarding	Amazon tools, GS1 DataKart
Pre-sale product Q&A;	Amazon's Rufus
Blind in-store shopping	Be My Eyes with Tesco
Screenplay coverage	Filmustage, Prescene
Flash-safe video cuts	Harding FPA; simple rules, little AI needed
Cultural checks for streaming	Spherex
Ad safety in videos	Vionlabs, Zefr
Podcast knowledge search	Snipd and others
Local sports highlights	CricHeroes already has AI highlights
Newsroom archive memory	Google Pinpoint and others
Children's picture books	Gemini Storybook already does it
TV ad-time logger (India)	The 12-minute ad limit was cancelled in September 2026
General AI time study (factory)	Coats, MTM, Retrocausal

10. Our recommendation

- **Best Retail idea:** SafeShelf. Serious problem, real AI work, and we can prove it with past data.
- **Best Media idea:** Akshara Ladder. Clearly new, big social impact, easy to measure.
- **Best Manufacturing idea:** PitStop. Simple story and clear before-and-after numbers.
- All three are roughly tied with our current plan, ClearSight. Switching only makes sense if we prefer one of these themes.
- **Suggested next step:** spend about 2 hours on the one key test for the chosen idea before deciding.

Remember: a well-built, well-presented idea beats a clever idea that is half finished.